

Cathy Liu

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EDUCATION

2024 – Jul 2026 Master of Biotechnology, **The University of Melbourne** (Grades: 74/100)
2020 – 2024 Bachelor of Science (Pharmaceutical Science), **Guangdong Medical University**
(Grades: 85/100)

WORK EXPERIENCE

Research Analyst

Feb 2025 – Dec 2025

– **Murdoch Children's Research Institute (MCRI)**

- Led Biotech Interventions workstream in an industry-partnered project on childhood malnutrition.
- Assessed effectiveness of biotechnology interventions for improving childhood malnutrition
- Synthesised biological readouts (growth indices, inflammatory markers, microbial signals) into decision-oriented technical summaries.
- Delivered technical briefs to mixed scientific and non-scientific stakeholders.

Research Assistant

Jul 2023 – Jun 2024

– **Beijing Institute of Biological Products Co., Ltd.**

- Contributed to early-stage R&D of ACYW135 quadrivalent meningococcal polysaccharide vaccine, spanning extraction, purification, formulation, and immunogenicity assessment.
- Developed and validated rocket electrophoresis as a QC method for polysaccharide content quantification, assessing specificity, reproducibility, linearity, and accuracy.
- Characterised polysaccharide molecular weight and purity via HPLC/SEC and agarose gel electrophoresis to support formulation optimisation.
- Conducted physicochemical stability testing and supported animal immunogenicity studies across all four serogroups; maintained GMP/SOP-compliant documentation throughout.

Pharmacy Assistant

May 2023 – Aug 2023

– **Guangdong Women and Children's Hospital**

- Rotated across inpatient pharmacy, IV room, outpatient pharmacy, and drug warehouse; operated automated dispensing, sorting, and packaging equipment.
- Conducted prescription review, medication verification, and patient counselling; managed drug cataloging, inventory, and insurance documentation.
- Therapeutic Drug Monitoring Lab: processed blood samples and operated automated biochemical analysers for pharmacokinetic monitoring.

Research Assistant

Mar 2022 – Apr 2023

– **Shanxi Zhendong Pharmaceutical Co., Ltd.**

- Managed stakeholder relationships with health providers and hospital pharmacies to support product adoption.
- Coordinated supply continuity and identified implementation bottlenecks to inform engagement strategies.

PROJECTS

LLM-Guided Minimax Optimisation for Epilepsy Neurostimulation — Computational Research

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Built an end-to-end pipeline combining RAG-augmented LLM literature mining (Claude API), The Virtual Brain digital-twin simulation, and minimax optimisation to find neurostimulation settings that protect the worst-responding patient rather than the cohort average; benchmarked against Bayesian optimisation across a 20-patient simulated cohort.

AI-Driven PDAC Target Discovery & Peptide Design — Computational Research Project

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Built an end-to-end target-discovery pipeline integrating scRNA-seq differential expression, LLM-assisted mechanistic graph construction, and dual-strategy knockout ranking (Boolean solver + Claude Opus ranker); pipeline converged on EFS–SRC axis as lead hypothesis, benchmarked against DepMap pancreatic dependency data. Designed candidate inhibitory peptide eSI-1 via AlphaFold 3 structural modeling and ProteinMPNN sequence optimization, generating an 8-sequence ensemble with ~1.0 improvement in ProteinMPNN score over the native EFS fragment.

Developing Strategies to Treat Childhood Malnutrition — Industry Project

Evaluated biotechnology strategies for childhood malnutrition treatment, integrating mechanistic rationale, scientific evidence, and development considerations into structured briefing summaries.

ACYW135 Quadrivalent Meningococcal Polysaccharide Vaccine — R&D Project

End-to-end vaccine R&D project spanning molecular characterisation, QC method development, formulation optimisation, and preclinical immunogenicity evaluation for a quadrivalent meningococcal polysaccharide vaccine (ACYW135).

Glioma Pharmacology Research — University Research Project

Constructed orthotopic and subcutaneous murine glioblastoma models via stereotaxic intracranial injection of U87MG-Luc cells; monitored tumor progression weekly via bioluminescence imaging (IVIS). Administered therapeutics via i.p., i.v., and p.o. routes across concurrent cohorts; performed endpoint collection of brain tissue, peripheral organs, and blood for integrated efficacy, PK, and safety assessment.

SKILLS

Laboratory Techniques	Mammalian cell culture, qPCR, RNA extraction, Western blot, ELISA, flow cytometry, fluorescence microscopy, plate reader assays (fluorescence, absorbance), HPLC/SEC, gel electrophoresis, rocket electrophoresis, sterile cell handling
Molecular & Analytical	Polysaccharide extraction/purification, physicochemical characterisation, method validation (specificity, linearity, accuracy), GMP/SOP compliance
Bioinformatics & Software	R, Python, MATLAB, GraphPad Prism, Microsoft Excel, Teselagen DNA design platform, MS Project
Communication	Technical writing, data reporting, cross-functional stakeholder communication
Languages	English (fluent), Mandarin Chinese (native)